

AI-POWERED SMART LIBRARY SYSTEM

Intelligent Automation for Modern School
Libraries



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INTRODUCTION & OBJECTIVES

School libraries are critical to student learning, yet most institutions still rely on manual, paper-based processes for book management, checkouts, and record-keeping. These outdated workflows lead to inefficiencies, errors, and reduced engagement. The AI-Powered Smart Library System addresses this challenge by automating school library workflows using Artificial Intelligence — enabling faster operations, smarter recommendations, and a significantly reduced administrative burden on library staff.



Streamline Book Allocation

Automate the process of issuing, returning, and tracking books to eliminate manual errors and reduce wait times for students and staff.



Dynamic Recommendation Models

Implement AI-driven recommendation algorithms that suggest personalized reading materials to students based on their reading history and preferences.



Reduce Manual Administrative Load

Minimize repetitive administrative tasks such as manual cataloguing, record updates, and inventory checks through intelligent automation and smart reporting.

AI SOLUTION & KEY FEATURES

Our solution integrates OCR vision systems with advanced Machine Learning algorithms to automate and enhance every aspect of the school library experience — from checkout to inventory management.



AUTOMATED CHECKOUT

OCR scanner enables fast, error-free book checkout without manual input.



Reading Recommendations

Personalized ML models suggest books based on student interests and history.



Smart Inventory

AI-driven inventory tracking reduces loss and optimizes book availability.

BUDGET ANALYSIS & BENEFITS

Estimated total project expense has been calculated across three key categories: Hardware, Software, and Setup. The AI-Powered Smart Library System delivers measurable value through fast checkouts, increased student engagement, and reduced manual record usage.



Hardware Costs

45%

Scanners, RFID tags, kiosks, and computing devices required for automated checkout and smart inventory management infrastructure.

Software & AI

35%

Machine learning models, OCR vision systems, recommendation engine licenses, and library management software integration costs.

Setup & Training

20%

Installation, system configuration, staff training, and initial onboarding to ensure smooth deployment and adoption across the library.

CONCLUSION

Future Scope: The AI-Powered Smart Library System can be further enhanced by integrating voice assistant support for hands-free access and facial recognition technology for secure, seamless student identification — improving accessibility and automating library entry and checkout processes.

Conclusion: AI has the transformative potential to fully modernize school library operations. By automating checkouts, personalizing recommendations, and reducing manual workloads, this system sets a new standard for smart, efficient, and student-centered library management at Aravali International School.



THANK YOU!



Questions & Discussion



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